■ Soil Health & Crop Recommendation Report

Generated on: November 06, 2025 at 20:10:50 UTC

Farmer Information	
Name:	nandish more
Phone:	8088117817
Location:	pune
Land Size:	100.00 acres

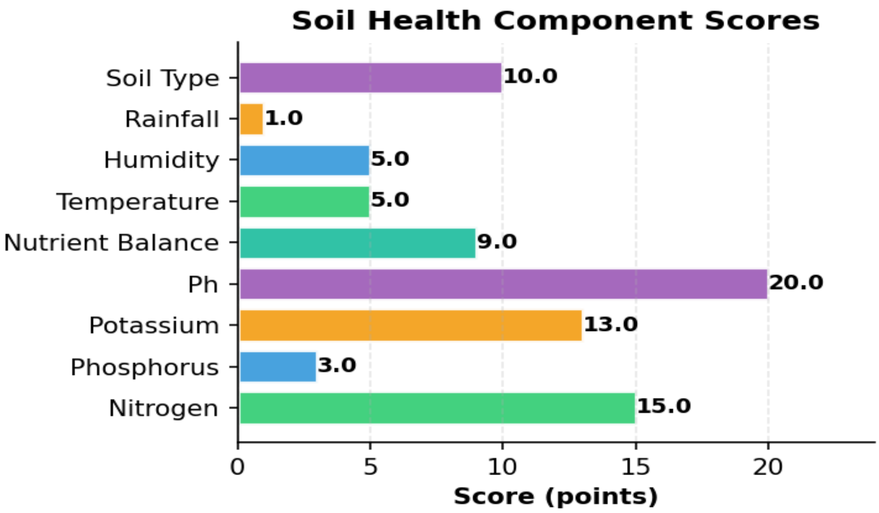
Soil & Weather Data

Parameter	Value	Unit
Nitrogen (N)	67.0	mg/kg
Phosphorus (P)	82.0	mg/kg
Potassium (K)	66.0	mg/kg
pH Level	6.50	
Temperature	29.9	°C
Humidity	75.9	%
Rainfall	12.0	mm
Soil Type	Alluvial Soil	

Soil Health Assessment

Overall Soil Health Score 	81.0/100	Excellent
Nitrogen	15.0	points
Phosphorus	3.0	points
Potassium	13.0	points
Ph	20.0	points
Nutrient Balance	9.0	points
Temperature	5.0	points
Humidity	5.0	points
Rainfall	1.0	points

Soil Type	10.0	points



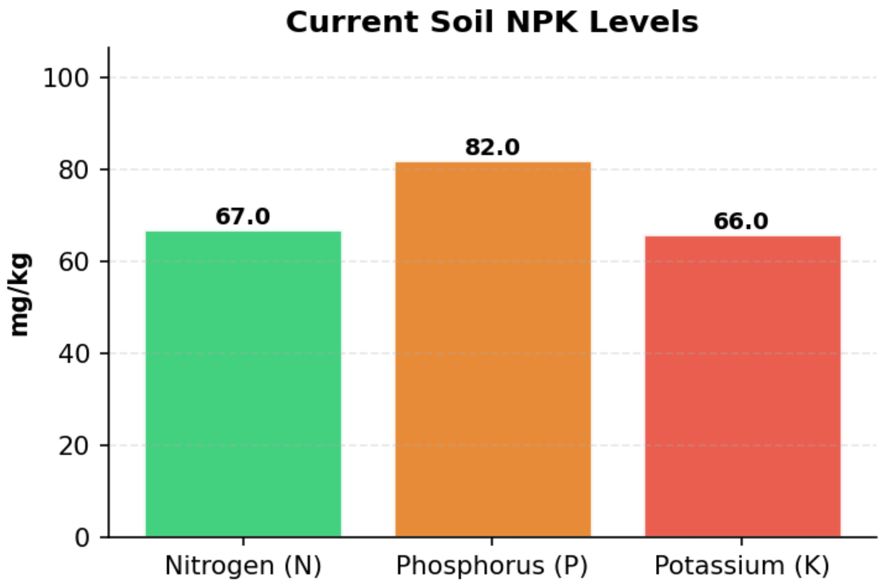
Recommended Crop

Predicted Crop	Muskmelon
Model Used	Stacking Classifier
Confidence Level	62.4%

Alternative Crop Options

Rank	Crop	Confidence
2	Banana	14.4%
3	Rice	1.5%

Soil NPK Levels Visualization



■■ Important Warnings

• Very high phosphorus levels detected. May interfere with micronutrient uptake.

Soil Treatment Recommendations

• Soil pH (6.5) is within optimal range (6.0-7.0) for muskmelon.

• Nitrogen (67.0 mg/kg) is within optimal range for muskmelon.

• Phosphorus (82.0 mg/kg) is excessive for muskmelon. Optimal range: 30-50. High P can interfere with micronutrient uptake.

• Potassium (66.0 mg/kg) is within optimal range for muskmelon.

• Rich in nutrients, good water retention. Suitable for most crops.

Fertilizer Recommendations

■ **Fertilizer Requirements for muskmelon:**

• Land Size: 100.0 acres

• Current NPK: N=67.0, P=82.0, K=66.0 mg/kg

• Ideal NPK: N=80, P=40, K=60 kg/acre

Required Application:

• Nitrogen (N): 1300.0 kg total (13.0 kg/acre)

→ Apply in 2-3 split doses: 50% at sowing, 25% at tillering, 25% at flowering

■ **Organic Alternatives:**

• Compost/FYM: 5-10 tons/acre (provides N, P, K)

• Green Manure: Leguminous crops (fixes N)

• Biofertilizers: Rhizobium (N), PSB (P), VAM (P)

Crop-Specific Tips for Muskmelon

Requires warm climate and well-drained soil. Provide support for vines.

Additional Notes

• These recommendations are based on comprehensive soil analysis and advanced ML predictions.
• Actual results may vary based on local conditions, weather patterns, and farming practices.
• Consult with local agricultural experts before making major farming decisions.
• Regular soil testing (every 2-3 years) is recommended for optimal crop management.
• Consider crop rotation and organic farming practices for long-term soil health.
• Monitor weather forecasts and adjust irrigation schedules accordingly.